

DATA423-26S1 Assignment 1

20% of your final grade

Summary

This assignment is about gaining ability and confidence in the use of Shiny. It is also about demonstrating the implementation of charts covered in labs. In particular, it should show the use of controls to add value to the charts. There is also some subtle aspects of the data to discover and discuss in the report.

This will test chart interpretation skills and report writing skills.

Part 1

Create a Shiny app using **RStudio**. Using the supplied comma separated variables (csv) file, **Ass1Data.csv**, load the data and use Shiny to summarise and visualise the information within.

There is no need to perform any modelling.

This deliverable should be submitted as a set of files (**ui.R**, **server.R** and possibly **global.R**) that we should be able to run and grade (without having to make any changes).

Recommendation: It is highly recommended to install the latest versions of the R packages you use.

Part 2

Using the supplied data and your app, write a report explaining what interesting features you have found in the data, support this with appropriate visualizations taken from your app.

Please ensure your name and student-id are provided in the document header or footer.

Hint: Take a look at the document “Phrases that might be applicable for your report”, available on LEARN in the Assessment section for Assignment 1.

Hint: Have a sub-heading for each plot. Your report should have at least the following sections:

- Introduce Data
- Mosaic Chart
- ggPairs Plot
- Correlation Plot
- Missingness
- Boxplot
- Rising Order Chart
- Datatable

Hint: All your figures and tables must be numbered and have a detailed description. Each figure or table must be referenced in your report. If not, why is it there?

Hint: Avoid using bullet points and numbered lists. Write using sentences and paragraphs.

Typically, reports will be 6-10 pages in length (including figures and tables).

This deliverable MUST be submitted as a PDF document.

Detail

Steps

1. Create a shell of a Shiny app. Plan whether you want a sidepanel/main layout or a fluidpage layout or something more ambitious. Design how the user would progressively get more information by interacting with the page. Try to avoid long pages - instead try to use a tabset or something similar to control your navigation through the charts.
2. Add your name to the title part of the **ui** so it is clear to see whose app is running. Something like:

```
...
title = "Assignment 1 - John Smith",
...
```

3. Place the CSV file in the same location as the ui.R, server.R files. Load the csv file using:

```
dat <- read.csv("Ass1Data.csv", header = TRUE, stringsAsFactors = TRUE)
```

4. Check the variable types that have been assigned. You will need to manually intervene for date types and for ordinal factors.
5. Provide several visualisations about the data set. As a minimum provide the following:
 - a) a **vcd::mosaic** plot of discrete variables
 - b) a **GGally::ggpairs** plot of all variables
 - c) a **corrgram::corrgram** correlation chart
 - d) a **visdat::vis_miss** showing missing data
 - e) a **DT::datatable** a listing of the data
 - f) a **car::Boxplot** of numeric variables (Avoid the **formula** version of the **car::Boxplot()** function as this is buggy)
 - g) a rising-value chart would help to detect gaps in the continuous data
6. In all visualisations, make sure that any relevant charting parameters have been provided as shiny controls. For example, box plots needs to respond to a slider that controls the threshold for marking points as outliers, and to check boxes for centring and scaling. We expect that most charts will need controls to manage their appearance.
7. Test your shiny app to ensure it will run on another machine, (use Session / Restart R and Session / Clear Workspace in the RStudio menu). Every year students submit assignments that will not run without the marker making changes - typically they forget to declare packages (i.e. **library(ggplot2)**) or use variables that are set up outside of the supplied code. This will lose you marks.
8. Write up what you have learned from the visualisations that was special / unusual / curious about this data set. Try to say something about each plot. For example, what does the **vis_miss** chart tell us about missing values? Convert this document to a **pdf**.
9. Use the course LEARN website to upload the report **pdf**. Compress the shiny files (**ui.R**, **server.R**, **global.R**) using zip and upload the resulting **zip** file.

Considerations

Data Types

Do the data types of any variables need to be programmatically altered? Are some strings actually dates? Are factors really ordered factors? Are there any cyclic variables?

Summary

Provide some summary statistics about the data set, `summary()`, `dfSummary()` or `tibble::glimpse()` can achieve this. In particular make sure that the number of variables and the number of observations are presented somewhere in your app.

Decorate

Decorate your visualisations with appropriate axis labels and a main title. Ensure all your visualisations have figure numbers and a detailed description. Having a figure number allows easy referral to your visualisations in your report.

Type-appropriate

When doing things like scaling:

```
data2 <- scale(data, center = input$center, scale = input$scale)
```

this will not work on non-numeric data. You will only need to scale the subset of columns of the data set that are numeric. This also applies to charts that only apply to particular data types e.g. discrete continuous numeric factor variables.

Interactive

Try to add one or more controls to each visualisation that will vary what information it shows and/or how it is shown.

Complexity

Avoid showing too much information in a single chart. However, if the user selects too many variables to display, that is their (poor) choice. Complex charts can take a long time to draw. Always set any controls to defaults that produce a simple & fast chart.

Marking

This assignment is worth 20% of your final grade.

We mark the assignment out of 100 with 50% of the mark for the Shiny app and 50% for the report.

If you use an LLM or equivalent to aid your report writing and/or coding please acknowledge this in your report.

We are looking for charts that reveal features of this data. The data has been synthetically constructed to have interesting features - can you find them all?

We are looking for charts that operate selectively on only those column types for which it is appropriate. For example box plots that do not plot the factor columns and scatter plots that do not plot the factor columns. Mosaic plots that only use discrete columns.

We are looking for charts that are not too complicated. Beware of trying to show all variables in the **ggpairs** plot. This will take an age and is clearly not workable. How can you sensibly divide the variables into sets that can be charted separately?

We are looking for charts that change as the user interacts with check boxes and sliders. Chart titles and subtitles should incorporate/reflect the options used to create the chart.

The text document should show consistent and correct thinking about outliers, missing data, and data roles.

We will reserve extra marks for something impressive in your shiny app although we will not know what that is until we see it.

Go on, impress us!

The End